



Higher  
Coursework  
Assessment Task



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# Higher Computing Science Assignment Finalised Marking instructions

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# Marking instructions

## General marking principles

This information is provided to help you understand the general principles that must be applied when marking candidate responses in this assignment. These principles must be read in conjunction with the specific marking instructions, which identify the key features required in candidate responses.

- a Marks for each candidate response must always be assigned in line with these general marking principles and the specific marking instructions for this assessment.
- b Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

## Specific marking instructions

| Task     | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|
| <b>1</b> | <b>Software design and development</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                |
| 1a       | <p>1 mark for each bullet. Max 2 marks.</p> <ul style="list-style-type: none"> <li>♦ date is correct (UK) format (day/month)</li> <li>♦ date is within an acceptable range e.g. no 31<sup>st</sup> September</li> <li>♦ age is a whole number/integer</li> <li>♦ age is within an acceptable range</li> <li>♦ only one town and one mammal on each line</li> </ul>                                                                                                                                                         | <p>For two marks, must have two bullets from left.</p> <p>Award 1 mark for general reference to data that is valid/present/complete/formatted.</p> <p>Must specify town and mammal or state 4 values per line.</p>                                                                                                                                                                                               | 2               | Analysis       |
| 1b       | <p>1 mark for each bullet.</p> <ul style="list-style-type: none"> <li>♦ identify first character</li> <li>♦ if not upper-case convert to upper-case</li> <li>♦ concatenate with remaining string</li> <li>♦ return string/variable/value</li> </ul>                                                                                                                                                                                                                                                                        | <p>Accept reference to pre-defined functions in design for conversion.</p> <p>Return value should be referenced in previous steps.</p>                                                                                                                                                                                                                                                                           | 4               | Design         |
| 1c       | <p>Read in Mammals Data (2)</p> <ul style="list-style-type: none"> <li>♦ module with correct parameter passed or returned to read data from file to array of records</li> <li>♦ each line of sightings data stored in a record structure assigned to an array</li> </ul> <p>Find oldest person (3)</p> <ul style="list-style-type: none"> <li>♦ module with correct parameter passed and max displayed within procedure</li> <li>♦ initialise and re-assign max age</li> <li>♦ if statement to find correct max</li> </ul> | <p>If candidate uses parallel arrays as a data structure, award 0 marks for “read” procedure then accept appropriate parameter passing for parallel arrays in the remainder of the procedures/functions.</p> <p>Award 0 marks for bullets 2 and 3 if a pre-defined function is used in implementation instead of a finding maximum algorithm.</p> <p>Accept finding position of max value to produce output.</p> | 15              | Implementation |

| Task | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Additional guidance                                                                                                                                                                                                                                                                                                 | Marks available |         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|
| 1    | <b>Software design and development</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                     |                 |         |
|      | <p>Upper-case function (3)</p> <ul style="list-style-type: none"> <li>♦ extract first character</li> <li>♦ if statement to convert to upper-case using ASCII/Char pre-defined function</li> <li>♦ return original or concatenated string</li> </ul> <p>Display dates of sightings (2)</p> <ul style="list-style-type: none"> <li>♦ module with correct parameter passed, correct use of the function and dates displayed within procedure</li> <li>♦ linear search to find sighting dates</li> </ul> <p>Display the number of sightings each day (3)</p> <ul style="list-style-type: none"> <li>♦ module with correct parameter passed, dayToCount (date) and count displayed within procedure</li> <li>♦ count initialised to 1 and incremented for a single date</li> <li>♦ dayToCount and count reset for each new date</li> </ul> <p>Implementation (2)</p> <ul style="list-style-type: none"> <li>♦ a single upper-case function called twice</li> <li>♦ modular (4 procedures and 1 function) and maintainable</li> </ul> | <p>Do not deduct marks for two separate functions for town and mammal at this step.</p> <p>Do not penalise if no step 4.12 or display for 30/09/2021.</p> <p>Maintainability should include evidence of meaningful identifiers, internal commentary, indentation and white space in the context of the program.</p> |                 |         |
| 1d   | <p>1 mark for each bullet. Max 2 marks.</p> <ul style="list-style-type: none"> <li>• watchpoint set on count variable</li> <li>• count increments by 1 (while date is 01/09/21) OR count is 6 when dayToCount changes to 02/09/21</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• watchpoint set on dayToCount</li> <li>• count should be 6 when dayToCount moves to 02/09/21</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                     | 2               | Testing |

| Task | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Additional guidance | Marks available |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|
| 1    | <b>Software design and development</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |                 |
| 1e   | <p>Efficiency<br/>1 mark for any bullet.</p> <ul style="list-style-type: none"> <li>• A single function can be used to check first characters of town and mammal</li> <li>• First character only changes if found to be lower-case</li> <li>• Accept explanation for inefficient</li> </ul> <p>Maintainability<br/>1 mark for any bullet.</p> <ul style="list-style-type: none"> <li>◆ linking modularity to maintainability e.g. sub procedures can be edited independently</li> <li>◆ local variables prevent clashes with variables in other parts of the code</li> </ul> |                     | 2               |
|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     | Evaluation      |

| Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Expected response                                                                                                                                                                                                         | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks available |                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Database design and development                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                |
| 2a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1 mark for each bullet.<br>♦ entity names, in the correct order<br>♦ correct number of instances<br>♦ adding the correct associations                                                                                     | Entities could be written in reverse order                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3               | Analysis       |
| <div><div><div>Entity Names</div><div><div>Planner</div><div>Route</div><div>Walk</div><div>Walker</div></div><div><div>Planner 1</div><div>Planner 2</div><div>Planner 3</div></div><div><div>Route 1</div><div>Route 2</div><div>Route 3</div><div>Route 4</div></div><div><div>Walk 1</div><div>Walk 2</div><div>Walk 3</div><div>Walk 4</div><div>Walk 5</div><div>Walk 6</div></div><div><div>Walker 1</div><div>Walker 2</div><div>Walker 3</div><div>Walker 4</div></div></div></div> <div><div>Entity Names</div><div><div>Walker</div><div>Walk</div><div>Route</div><div>Planner</div></div><div><div>Walker 1</div><div>Walker 2</div><div>Walker 3</div><div>Walker 4</div></div><div><div>Walk 1</div><div>Walk 2</div><div>Walk 3</div><div>Walk 4</div><div>Walk 5</div><div>Walk 6</div></div><div><div>Route 1</div><div>Route 2</div><div>Route 3</div><div>Route 4</div></div><div><div>Planner 1</div><div>Planner 2</div><div>Planner 3</div></div></div> |                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                |
| 2b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1 mark for each bullet.<br>♦ fields, function, alias<br>- forename, surname, plannerNo<br>- COUNT of records with alias<br>♦ equijoins<br>♦ GROUP BY (forename), (surname), plannerNo<br>♦ ORDER BY COUNT of records DESC | <div>SELECT Planner.forename,<br/>Planner.surname, Planner.plannerNo,<br/>COUNT(Walk.walkID) AS [Total<br/>Participants]<br/>FROM Walk, Route, Planner<br/>WHERE Walk.routeID = Route.routeID<br/>AND Route.plannerNo =<br/>Planner.plannerNo<br/>GROUP BY Planner.forename,<br/>Planner.surname, Planner.plannerNo<br/>ORDER BY COUNT(Walk.walkID) DESC;</div> <div>Count can be on any field or *<br/>SQL allows ordering on alias for Count.</div> <div>4 records produced</div> | 4               | Implementation |

| Task     | Expected response                                                                                                                                                                                                                                                                                                                                                                                 | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks available |                |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|
| <b>2</b> | <b>Database design and development</b>                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |                |
| 2c       | 1 mark for each bullet. <ul style="list-style-type: none"> <li>◆ query to find longest walk using Max function</li> <li>◆ correct fields and tables</li> <li>◆ equi-joins</li> <li>◆ use of Query 1 result in the criteria to produce correct output</li> <li>◆ use of GROUP BY walkerNo (forename, surname, telNo) to remove walkers who have walked the longest route more than once</li> </ul> | Query using MAX function could be a sub-query within the criteria.<br><br>SELECT MAX(distance) AS longest<br>FROM Route;<br>(query saved as LongestWalk)<br><br>SELECT Walker.walkerNo,<br>Walker.forename, Walker.surname,<br>Walker.telNo<br>FROM Walker, Walk, Route,<br>LongestWalk<br>WHERE Walk.routeID = Route.routeID<br>AND Walker.walkerNo = Walk.walkerNo<br>AND distance = longest<br>GROUP BY Walker.walkerNo,<br>Walker.forename, Walker.surname,<br>Walker.telNo;<br><br>Award 0 marks for bullet 3 if innerjoin is used<br>Award 0 marks for bullet 4 if value is used instead of field. | 5               | Implementation |
| 2d       | 1 mark for each bullet. <ul style="list-style-type: none"> <li>◆ use of wildcard before shoe/s</li> <li>◆ printed evidence of new SQL statement accommodating additional types of shoe producing same output</li> </ul>                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2               | Testing        |
| 2e       | ◆ find the travelling distance from a walker's home to the starting point of a chosen route<br>OR<br>◆ display walkers who prefer to walk a route with a chosen level of difficulty                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1               | Evaluation     |

| Task     | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Additional guidance                                                                                                                                                             | Marks available |          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|
| <b>3</b> | <b>Web design and development</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                 |                 |          |
| 3a       | <p>End user<br/>1 mark for any bullet. Max 2 marks.</p> <ul style="list-style-type: none"> <li>◆ read/find/get information about woodland walks</li> <li>◆ read/buy maps (with route directions)</li> <li>◆ read/get information about local wildlife</li> <li>◆ read/get information about lengths of walks</li> <li>◆ read/get information about the rating of walks</li> <li>◆ book walking tours online</li> </ul> <p>Functional<br/>1 mark for any bullet. Max 1 mark.</p> <ul style="list-style-type: none"> <li>◆ order form/link to buy maps.</li> <li>◆ form/link to book walking tours online.</li> <li>◆ navigation bar to move between pages of the website</li> <li>◆ formatted web pages displaying text and graphics (in context)</li> </ul> | <p>Must indicate a feature of the software.</p>                                                                                                                                 | 3               | Analysis |
| 3b       | <p>1 mark for each bullet.</p> <ul style="list-style-type: none"> <li>◆ (annotated) wireframe showing walks, description and map</li> <li>◆ description of show/reveal when walk element is selected</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>There may be a variety of approaches here but any solution should limit info to a page but still retain a method by which an element will show/reveal walks individually</p> | 2               | Design   |



| Task     | Expected response                                                                                                                                                                                                                                                                                                     | Additional guidance                                                                                | Marks available |                |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------|----------------|
| <b>3</b> | <b>Web design and development</b>                                                                                                                                                                                                                                                                                     |                                                                                                    |                 |                |
| 3c       | 1 mark for each bullet. <ul style="list-style-type: none"> <li>◆ IDs assigned to sections (walks)</li> <li>◆ Walk 2 and Walk 3 sections set to display:none ;</li> <li>◆ onmouseover events added to h3 elements displayed on one line</li> <li>◆ functions displaying one element while hiding the others</li> </ul> |                                                                                                    | 4               | Implementation |
| 3d       | 1 mark for each bullet. <ul style="list-style-type: none"> <li>◆ creating sections/divs for each resized image and description</li> <li>◆ use of float to position graphics and text to right or left as per design</li> <li>◆ addition of Gallery link to nav bar</li> </ul>                                         | Accept various implementations of float attribute<br><br>Gallery link not required on other pages. | 3               | Implementation |
| 3ei      | 1 mark for each bullet. <ul style="list-style-type: none"> <li>◆ test should include no selection AND multiple selections of a wood</li> <li>◆ test should include inputs within a range 0 to 5 AND outwith range 0 to 5</li> </ul>                                                                                   | Exceptional input should not be text as the web page does not allow this to be entered.            | 2               | Testing        |
| 3eii     | 1 mark for any bullet. Max 1 mark. <ul style="list-style-type: none"> <li>◆ lack of validation on text fields (invalid contact emails etc)</li> <li>◆ form can be submitted without a wood selected</li> <li>◆ user cannot book multiple walks on the same form</li> </ul>                                            |                                                                                                    | 1               | Evaluation     |

| Task 1 - Software Design and Development |                                                                    | Marks Available | Marks Awarded |
|------------------------------------------|--------------------------------------------------------------------|-----------------|---------------|
| 1a - Analysis                            | Assumptions                                                        | 2               |               |
|                                          |                                                                    |                 | <b>/2</b>     |
| 1b - Design                              | Identify first character                                           | 1               |               |
|                                          | Upper-case conversion                                              | 1               |               |
|                                          | Concatenation                                                      | 1               |               |
|                                          | String/variable returned                                           | 1               |               |
|                                          |                                                                    |                 | <b>/4</b>     |
| <b>1c - Implementation</b>               |                                                                    |                 |               |
| Read in mammals data                     | Module with parameter                                              | 1               |               |
|                                          | Sightings data stored in record structure and assigned to an array | 1               |               |
| Find oldest person                       | Module with parameter                                              | 1               |               |
|                                          | Initialise and re-assign max age                                   | 1               |               |
|                                          | IF statement                                                       | 1               |               |
| Upper-case function                      | Extract first character                                            | 1               |               |
|                                          | IF statement                                                       | 1               |               |
|                                          | Return original or concatenated string                             | 1               |               |
| Display dates of sightings               | Module with parameter                                              | 1               |               |
|                                          | Linear search                                                      | 1               |               |
| Display number of sightings each day     | Module with parameter                                              | 1               |               |
|                                          | Count initialised and incremented                                  | 1               |               |
|                                          | dayToCount and count reset                                         | 1               |               |
| Implementation                           | Single upper-case function called twice                            | 1               |               |
|                                          | Modular and maintainable                                           | 1               |               |
|                                          |                                                                    |                 | <b>/15</b>    |
| 1d - Testing                             | Watchpoint set                                                     | 1               |               |
|                                          | Counted correctly                                                  | 1               |               |
|                                          |                                                                    |                 | <b>/2</b>     |
| 1e - Evaluation                          | Efficiency of function                                             | 1               |               |
|                                          | Maintainability - modularity                                       | 1               |               |
|                                          |                                                                    |                 | <b>/2</b>     |

| Task 2 - Database Design and Development |                            | Marks Available | Marks Awarded |
|------------------------------------------|----------------------------|-----------------|---------------|
| 2a - Design                              | Entities                   | 1               |               |
|                                          | Instances                  | 1               |               |
|                                          | Associations               | 1               |               |
|                                          |                            |                 | <b>/3</b>     |
| 2b- Implementation                       | Fields, function and alias | 1               |               |
|                                          | Equi joins                 | 1               |               |
|                                          | Group by                   | 1               |               |
|                                          | Order by                   | 1               |               |
|                                          |                            |                 | <b>/4</b>     |
| 2c - Implementation                      | Query using Max function   | 1               |               |
|                                          | Fields and tables          | 1               |               |
|                                          | Equi-joins                 | 1               |               |
|                                          | Use of Query 1             | 1               |               |
|                                          | Group by                   | 1               |               |
|                                          |                            |                 | <b>/5</b>     |
| 2d - Testing                             | Use of wildcard            | 1               |               |
|                                          | Alias/printout             | 1               |               |
|                                          |                            |                 | <b>/2</b>     |
| 2e - Evaluation                          | Functional requirement     | 1               | <b>/1</b>     |

| Task 3 - Web Design and Development |                                   | Marks Available | Marks Awarded |
|-------------------------------------|-----------------------------------|-----------------|---------------|
| 3a - Analysis                       | End-user requirements             | 2               |               |
|                                     | Functional requirement            | 1               |               |
|                                     |                                   |                 | <b>/3</b>     |
| 3b - Design                         | Wireframe                         | 1               |               |
|                                     | Description of show/reveal        | 1               |               |
|                                     |                                   |                 | <b>/2</b>     |
| 3c - Implementation                 | IDs assigned to sections          | 1               |               |
|                                     | Walks 2 & 3 sections display none | 1               |               |
|                                     | Onmouseover events on h3 element  | 1               |               |
|                                     | Function(s)                       | 1               |               |
|                                     |                                   |                 | <b>/4</b>     |
| 3d - Implementation                 | Sections/divs                     | 1               |               |
|                                     | Float                             | 1               |               |
|                                     | Gallery link                      | 1               |               |
|                                     |                                   |                 | <b>/3</b>     |
| 3e(i)- Testing                      | None and multiple selections      | 1               |               |
|                                     | Range of numbers                  | 1               |               |
|                                     |                                   |                 | <b>/2</b>     |
| 3e(ii) - Evaluation                 | Form evaluation                   | 1               | <b>/1</b>     |

[END OF MARKING INSTRUCTIONS]